

MUHAMMAD GEMILANG RAMADHAN

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Summary

I am a final-year Electrical Engineering student at ITS Surabaya, passionate about data science and machine learning. Proven track record designing and deploying deep learning models, ranging from EfficientNet-based computer vision applications to reinforcement-learning based drone navigation using Python. Skilled in web development, data analytics, and fluent in both English and Indonesian. Passionate about leveraging AI and clean code to solve real-world problems and drive innovation.

Skills

- **Data & ML:** Python, NumPy, Pandas, TensorFlow, PyTorch, SQL, Excel, MATLAB.
- **Web Development:** HTML, CSS, JavaScript, React, Tailwind, PHP, Next JS, Express JS.
- **Version Control & Others:** Git, GitHub, Jira, Notion.

In addition, I understand how to prompt AI efficiently for greater use at problem solving. Fluent in both Indonesian and English to support in a team projects.

Education and Certification

- **Bachelor Program (expected to graduate in late 2025)**
Bachelor's degree – B.S., Electrical Engineering. Institute Technology of Sepuluh Nopember Surabaya.
- **Bangkit Machine Learning Path 2024 Graduates**
I graduated as a Machine Learning developer and got my ML and TensorFlow certification. Created a lot of machine learning projects such as mastering the unsupervised and supervised learning, OpenCV for image segmentation, and pattern recognition.
- **Udemy - 100 Days of Python**
Course: A bunch of series of Python programming courses, including projects, basic functions, development, etc.
- **MySkill - Ms. Excel in MySkill.**
Course: Mastering basic skills of Excel including pivot table, power pivot, and data visualization.
- **SQL for Data Analysis in MySkill.**
Course: Learning the fundamentals of SQL that is required for data analysis, manipulating table, etc.
- **Basic MATLAB**
Course: Learning the basics of MATLAB programming language.

Projects

- **Kulkas Babe's website (full stack e-commerce)**
A team of 2 where im focused on the front-end side, worked on a Laravel 11 multi-tenant e-commerce platform for a multi-outlet F&B delivery service, using Laravel Blade, Alpine.js, Tailwind CSS, Livewire, and vanilla JavaScript to build responsive, dynamic shopping experiences. Developed and optimized cart, checkout, and payment flows, integrated with external APIs (Google Maps for delivery fees, Midtrans for QRIS payments, Olsera POS for product data), and implemented caching strategies for performance. Collaborated closely with back-end services and followed structured development patterns to ensure maintainable, scalable, and secure UI components.
- **JejakBatik's application (.app)**
Consist a team of cloud, ML, and android dev to create a fully functional Batik pattern recognition as a final project of Bangkit program. I created a neural network architecture to recognize it by leveraging the EfficientNetB0 as the base model. Refer to my GitHub for further information.
- **Autonomous drone navigation and obstacle avoidance using Deep Deterministic Policy Gradient (DDPG)**
Creating my final year project of drone navigation and obstacle avoidance using actor-critic network. Designing and implementing stablebaselines3 DDPG, a reinforcement-learning based model into AirSim simulator for drone. Achieved model of ~80% accuracy where drone has high reward and accuracy in evaluation. Able to maneuver to avoid obstacles and reaches the goal efficiently.
- **Created website for my campus organization called Anargya ITS EV Team**
As a web developer role at Anargya ITS EV Team, I've created a front-end website where I leverage design from Figma. I was also assigned as a designer. So simultaneously, I'm a web developer and designer for Instagram feeds. Anargya ITS EV Team won 1st place as Overall Winner Pi-EV FSAE Japan.